

Mathematical Foundations of Political Methodology Lecture 5

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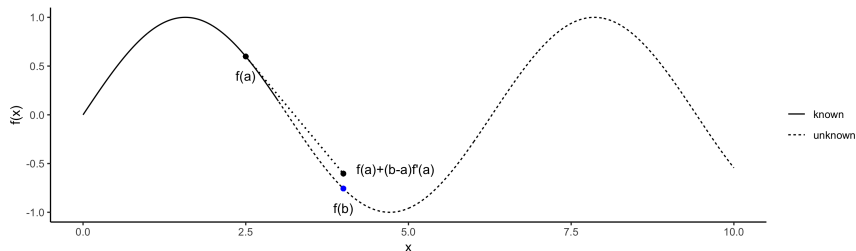
August 28th, 2020

- Last time we learned about the Mean Value Theorem.
- We will briefly return to that topic before turning to techniques for finding maxima and minima.
- How much can we infer about the value of a function at a point x given the curvature at a point x_0 ?

Linear Approximation.

- We want to know $f(b)$, we know $f(a)$, $f'(a)$
- With the slope and the distance between a and b we can calculate the position of $f(b)$:

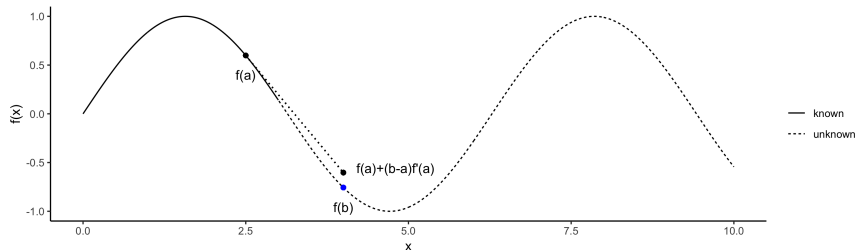
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- Can we do better?

Preliminary Result: Second Mean Value Theorem (pg 189).

Theorem

If f is a twice-differentiable function and a and b are real numbers such that $a < b$, there exists d such that $a < d < b$ and

$$f(b) = f(a) + (b - a)f'(a) + \frac{1}{2}(b - a)^2 f''(d)$$

Applying the Second MVT

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Set $b = a + h$, $d = a + th$ for some $0 < t < 1$:

$$f(a + h) = f(a) + (h)f'(a) + \frac{1}{2}(h)^2 f''(a + th)$$

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Replace $h = x - a$,

$$\tilde{f}(x) = f(a) + (x - a)f'(a) + \frac{1}{2}(x - a)^2 f''(a)$$

$\tilde{f}(x)$ is called the quadratic approximation of $f(x)$

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- If a function has arbitrary many higher derivatives, it is called smooth.
- We use these higher derivatives to approximate a function to arbitrary precision.

Taylor's Theorem

If f is a smooth function and n is a positive integer:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + \frac{h^n}{(n)!}f^{(n)}(a+th)$$

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The $(n-1)$ th order Taylor approximation sets the n th term to 0:

$$f(a+h) \approx f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \dots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + 0$$

Taylor's Series Expansion

In general, the Taylor's Approximations take advantage of Taylor's Series:

$$f(a + h) = f(a) + hf'(a) + \frac{h^2}{2}f''(a) + \frac{h^3}{3!}f'''(a) + \dots$$

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$$e^{0+h} = 1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} \dots$$

Binomial Series

Binomial Series

- We saw that e^x can be expanded into something that multiplies $x^0, x^1, x^2 \dots$,
- the book shows the same is true for $\ln(1+x)$.
- An important example of these power series is:

$$f(x) = (1+x)^c$$

- This formula gives rise to the *Binomial Series*.

Binomial Series $f(x) = (1 + x)^c$

$$f(x) = (1 + x)^c$$

$$f'(x) = c(1 + x)^{c-1}$$

$$f''(x) = c(c - 1)(1 + x)^{c-2}$$

$$f'''(x) = c(c - 1)(c - 2)(1 + x)^{c-3} \dots$$

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Consider $f(0)$:

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$$f'(0) = c(1)^{c-1}$$

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$$f'(0) = c$$

$$f''(0) = c(c - 1)$$

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Call $a_1 = c$, $a_2 = \frac{c-1}{2}$, $a_3 = \frac{c-2}{3} \dots a_k = \frac{c-(k-1)}{k}$

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$$f(0) = 1$$

$$f'(0) = c = a_1$$

$$f''(0) = c(c-1) = 2 * 1 * a_1 a_2$$

$$f'''(0) = c(c-1)(c-2) = 3 * 2 * 1 * a_1 a_2 a_3 \dots$$

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$$(1+x)^c = 1 + a_1x + \frac{2 * a_1a_2}{2!}x^2 + \frac{3!a_1a_2a_3}{3!}x^3 \dots$$

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$$a_{m+1} = \frac{m - m}{m + 1} = 0$$

$$(1 + x)^c = 1 + a_1x + a_1a_2x^2 + a_1a_2a_3x^3 \dots$$

$$(1 + x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-3)}{3!}x^3 + \dots + 1 * x^m + 0$$

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$$(1 + x)^m = \binom{m}{0}x^0 + \binom{m}{1}x + \binom{m}{2}x^2 + \binom{m}{3}x^3 + \dots + \binom{m}{r}x^r + \dots + \binom{m}{m}x^m$$

where:

$$\binom{m}{r} = \frac{m(m-1)(m-2)\dots(m+1-r)}{r!} = \frac{m!}{r!(m-r)!}$$

Binomial Series Example: $f(x) = (1 + x)^5$

$$(1 + x)^5 = \binom{5}{0}x^0 + \binom{5}{1}x + \binom{5}{2}x^2 + \binom{5}{3}x^3 + \binom{5}{4}x^4 + \binom{5}{5}x^5$$

$$(1 + x)^5 = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5$$

Pascal's Triangle

$n = 0$						$\binom{0}{0}$						
$n = 1$					$\binom{1}{0}$		$\binom{1}{1}$					
$n = 2$				$\binom{2}{0}$		$\binom{2}{1}$		$\binom{2}{2}$				
$n = 3$			$\binom{3}{0}$		$\binom{3}{1}$		$\binom{3}{2}$		$\binom{3}{3}$			
$n = 4$		$\binom{4}{0}$		$\binom{4}{1}$		$\binom{4}{2}$		$\binom{4}{3}$		$\binom{4}{4}$		
$n = 5$		$\binom{5}{0}$		$\binom{5}{1}$		$\binom{5}{2}$		$\binom{5}{3}$		$\binom{5}{4}$		
$n = 6$	$\binom{6}{0}$		$\binom{6}{1}$		$\binom{6}{2}$		$\binom{6}{3}$		$\binom{6}{4}$		$\binom{6}{5}$	$\binom{6}{6}$

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$n = 0$						1							
$n = 1$					1		1						
$n = 2$				1		2		1					
$n = 3$			1		3		3		1				
$n = 4$			1		4		6		4		1		
$n = 5$		1		5		10		10		5		1	
$n = 6$	1		6		15		20		15		6		1

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- Second Derivative Test
- Global Maxima
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- Strictly increasing: If $x_2 > x_1$ then $f(x_2) > f(x_1)$
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- Strictly decreasing: If $x_2 > x_1$ then $f(x_2) < f(x_1)$
- If $f'(x) > 0$ for all x , then f is strictly increasing; however this is stronger than we need for a function to be strictly increasing (e.g. $f(x) = x^3$)
- If $f'(x) \geq 0$ for all x and $f'(x) = 0$ for only a finite number of x , then f is strictly increasing (with strictly decreasing proved similarly)

Inverse Functions

Definition

Let $f(x)$ be monotonic. Then for any y , $f(x) = y$ has at most one solution. If the solution is written $x = g(y)$, then g is the inverse function of f , or

$$f(g(y)) = y \text{ and } g(f(x)) = x$$

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The inverse function is a reflection of the function about the 45° line; what's the inverse of the function $\{(1, 2), (2, 3), (3, 4)\}$?

The inverse function rule

If f is monotonic with inverse function g , and for a given x , f is differentiable and $f'(x) \neq 0$, then

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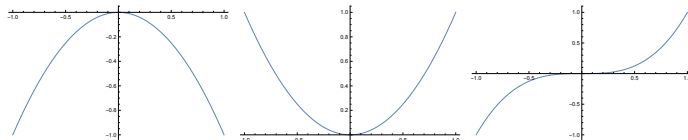
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This gives us another technique for differentiating things; we could calculate f' directly, or we could calculate $\frac{1}{g'}$

Critical Points

If $f'(x) = 0$ then x is called a *critical point*

- Maximum
- Minimum
- Inflection point



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- 4 Always remember that without more knowledge of f , all we know is that x is a *local* maximum or minimum (or an inflection point); we can't assume that it maximizes or minimizes our function globally

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- Second Derivative Test
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- In all of these cases, f lies above its tangent

Second Derivative Test

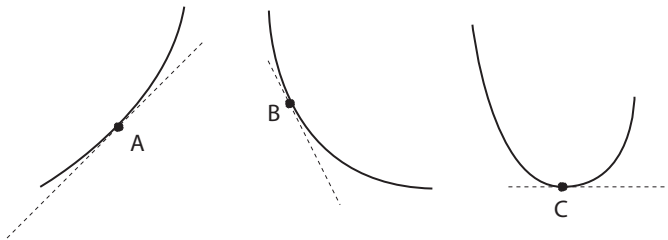
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Find and classify the critical points of $x^3 - 3x$, and then use this information to plot out the function

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- Convex and Concave Functions

Global Maxima

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Theorem

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- An important strategy is to trace out functions.

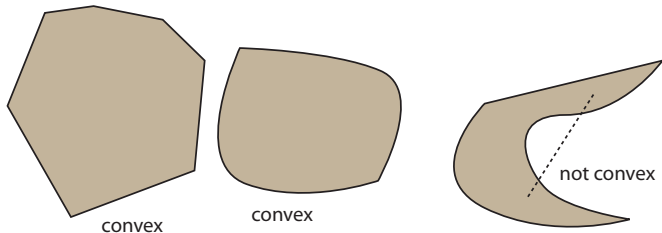
Trace out: $y = x - \frac{1}{x}$

- Monotone and Inverse Functions
- Second Derivative Test
- Global Maxima
- Convex and Concave Functions

Convexity and Concavity

Definition

A set is convex if the line segment connecting any two points in the set is entirely contained within the set

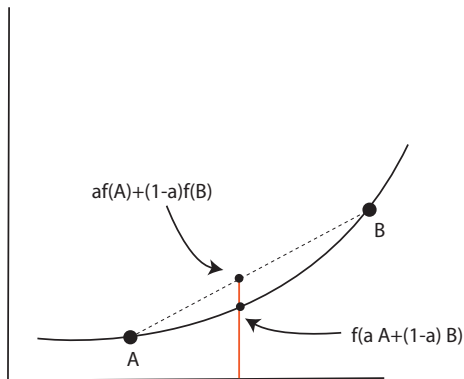


Convex Function

Definition

f is a convex function if and only if for all real numbers A, B and all $0 \leq \alpha \leq 1$,

$$\alpha f(A) + (1 - \alpha)f(B) \geq f(\alpha A + (1 - \alpha)B)$$



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f is concave if and only if $-f$ is convex: the set of all points on or below f is convex

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- Concave functions are typical choices for utility functions.

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- To remember, **concave** looks like the entrance to a cave.

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- Finally, check boundary points, and determine global maxima or minima.

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$f''(x) < 0$, local maximum

Example 1: $f(x) = -x^2$, $x \in [-3, 3]$

3) Check end points

$$\begin{aligned} f(0) &= -0^2 = 0 \\ f(-3) &= -(-3)^2 = -9 \\ f(3) &= -(3)^2 = -9 \end{aligned}$$

Example 2: $f(x) = x^3$, $x \in [-3, 3]$

1) Critical Values:

$$\begin{aligned}f'(x) &= 3x^2 \\0 &= 3(x^*)^2 \\x^* &= 0\end{aligned}$$

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2) Second Derivative:

$$\begin{aligned}f''(x) &= 6x \\ f''(0) &= 0\end{aligned}$$

No information

Example 2: $f(x) = x^3$, $x \in [-3, 3]$

3) Check End Points:

$$f(0) = 0^3 = 0$$

$$f(-3) = -3^3 = -27$$

$$f(3) = 3^3 = 27$$

Neither maximum nor minimum, saddle point

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We call μ legislator i 's **ideal point**

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$$\begin{aligned}U_i(\mu) &= -(\mu - \mu)^2 = 0 \\U_i(\mu - 2) &= -(\mu - 2 - \mu)^2 = -4 \\U_i(\mu + 2) &= -(\mu + 2 - \mu)^2 = -4\end{aligned}$$

Maximize utility at μ